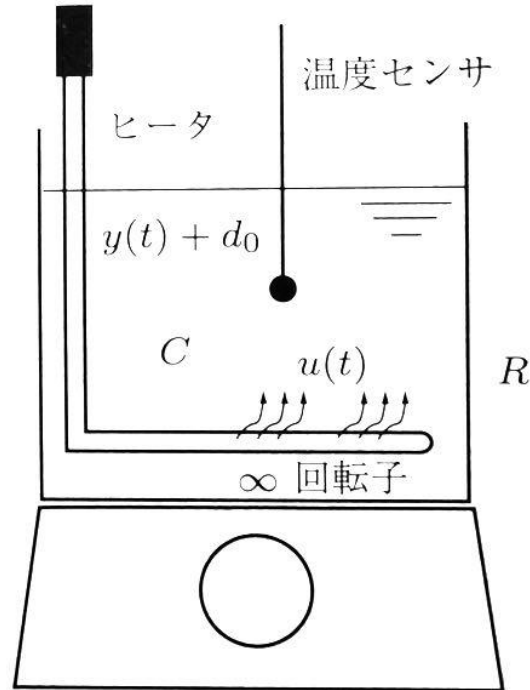
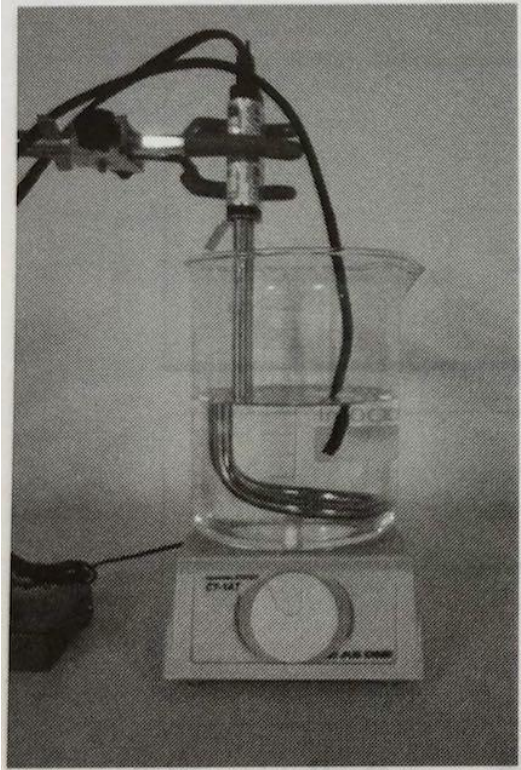

Step Response Test

— Electric, Electronic and System —
Engineering

Water Temperature Control (1/2)

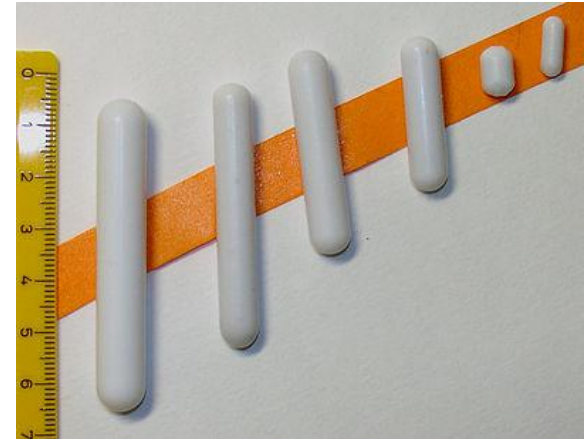
Setup



マグネチックスターラー

Magnetic Stirrer

a stir bar spins in a liquid
by magnetic field



Magnetic Stirrer

Inexpensive DOY Magnetic
Stirrer



Equation


$$\frac{d\tilde{y}(t)}{dt} = u(t) - \frac{\tilde{y}(t) - d_0}{R}$$


$$\frac{dy(t)}{dt} = u(t) - \frac{y(t)}{R}$$

C [J/K] : Heat Capacitor

R [K/W] : Heat Resistance

d_0 [K] : Outside Temperature

$u(t)$ [W] : Heat Input from Heater

$\tilde{y}(t)$ [K] : Water Temperature

$y(t)$ [K] : Difference between
Water and Outside Temperature

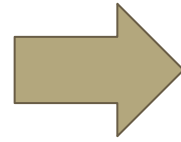
Laplace Transform ($y(0)=0$)

$$CsY(s) = U(s) - \frac{1}{R}Y(s)$$

$K=R$: Gain

$T=RC$: Response Time

$$\begin{aligned} Y(s) &= \frac{1}{\frac{1}{R} + Cs} U(s) \\ &= \frac{R}{1 + RCs} U(s) \end{aligned}$$



$$G(s) = \frac{K}{1 + Ts}$$

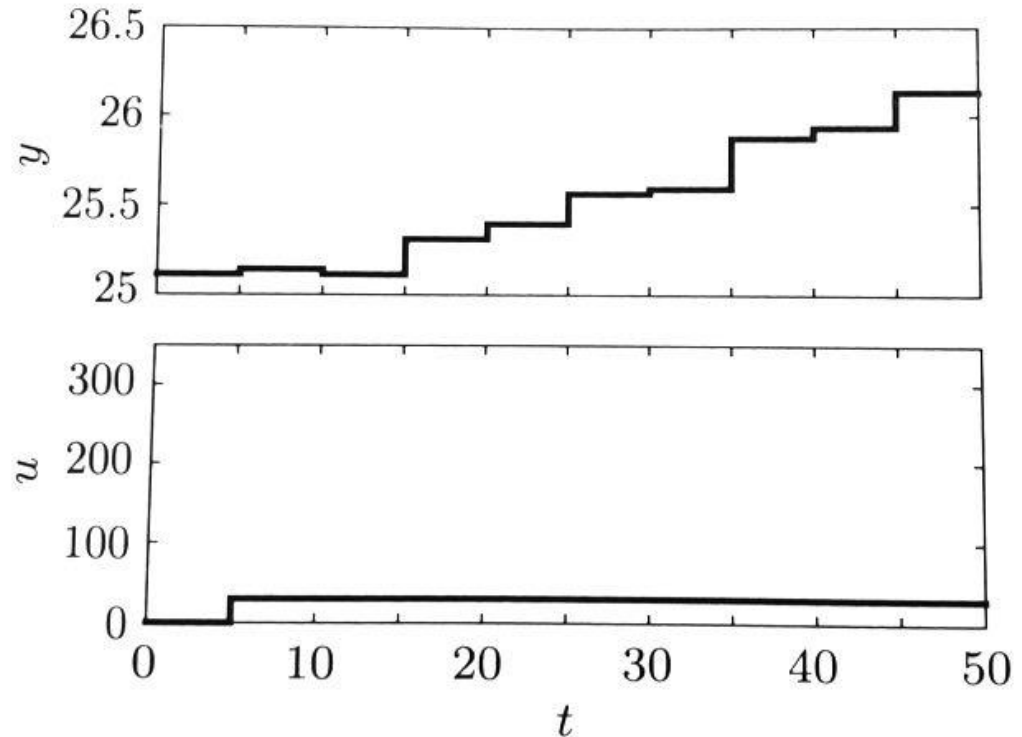
Identification of Parameters

Observe Step Input Response

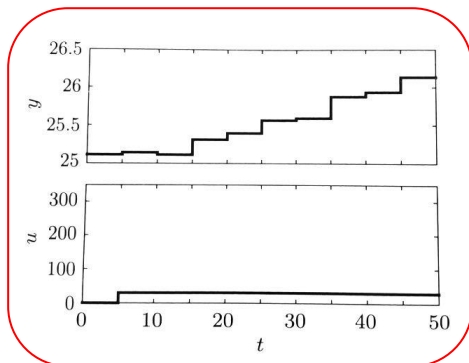
→ This system has 10-15s delay.

Plant $G(s)$ is **first order delay with dead time L**

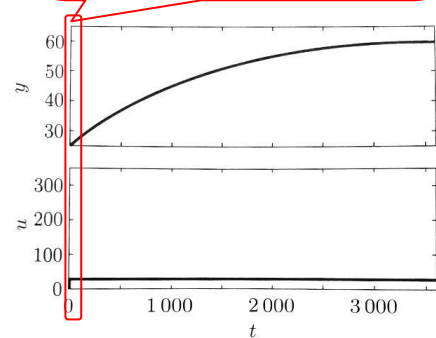
$$G(s) = \frac{K}{1 + Ts} e^{-Ls}$$



Identification from Response Curve

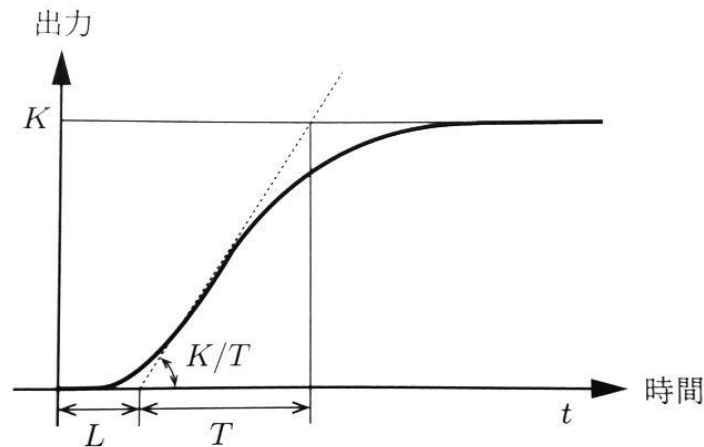


L →



K

T ↗



$$K = 1.15 \text{ K/W}$$

$$L = 15 \text{ s}$$

$$T = 1115 \text{ s}$$

System Identification

Input:

- Step
- Ramp
- Sine Wave
- Maximum-length sequence

Observe:

- Time-Domain
- Frequency-Domain

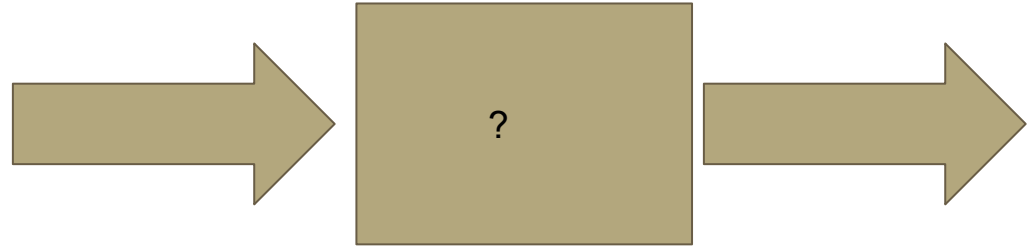


Image of System Identification

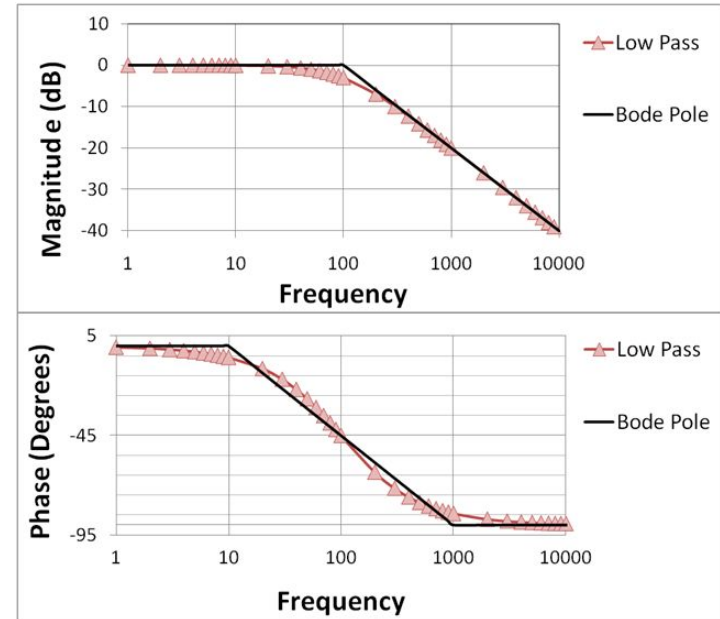
$$\frac{Y(s)}{U(s)} = \frac{K}{1 + Ts}$$

System Analysis

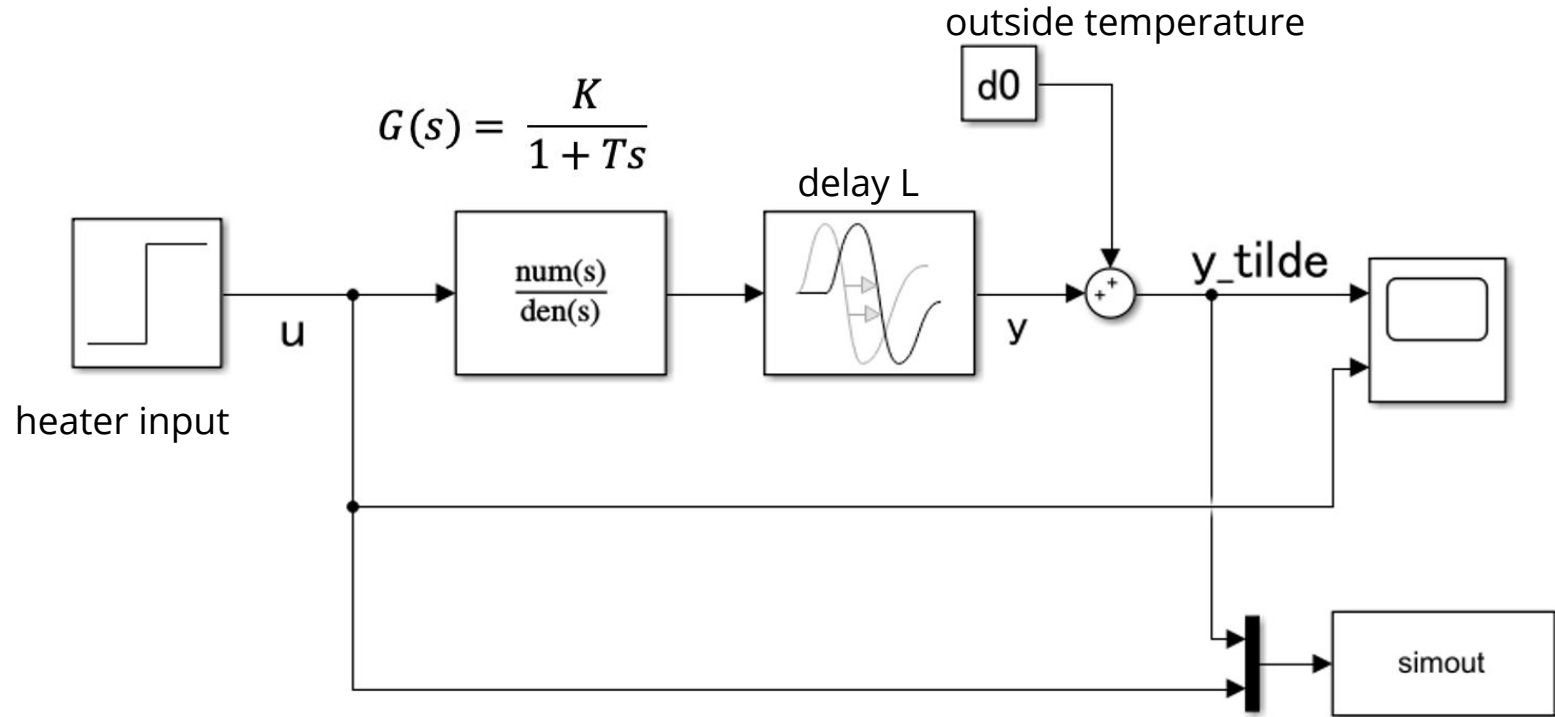


?

System Identification



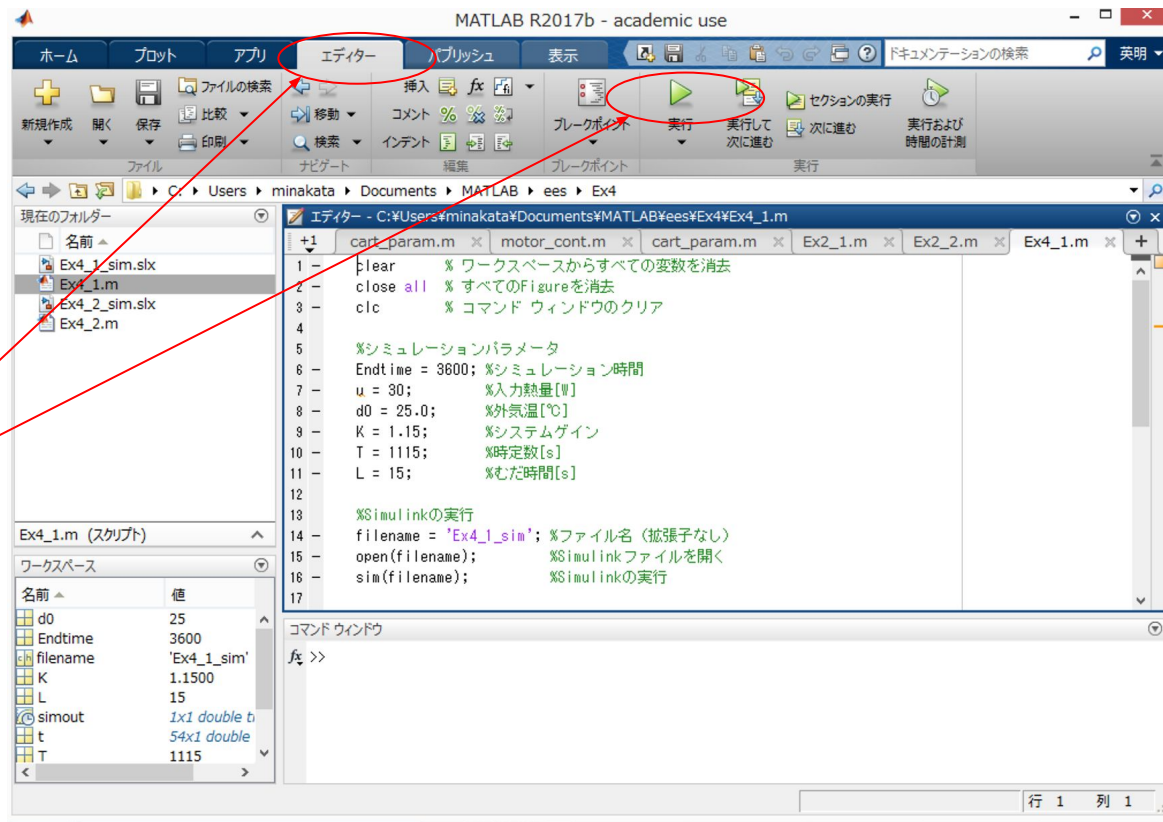
Implement Simulink Model



Control Simulations using m.file

Ex4_1.m controls
simulation named
Ex4_1_sim

1. open 'Ex4_1.m'
2. move to 'editor' tab
3. click 'run'

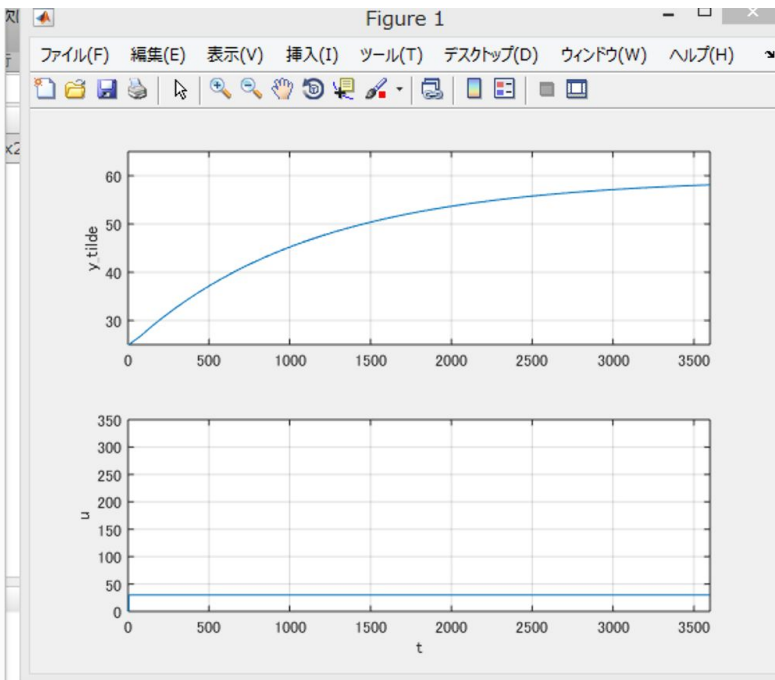
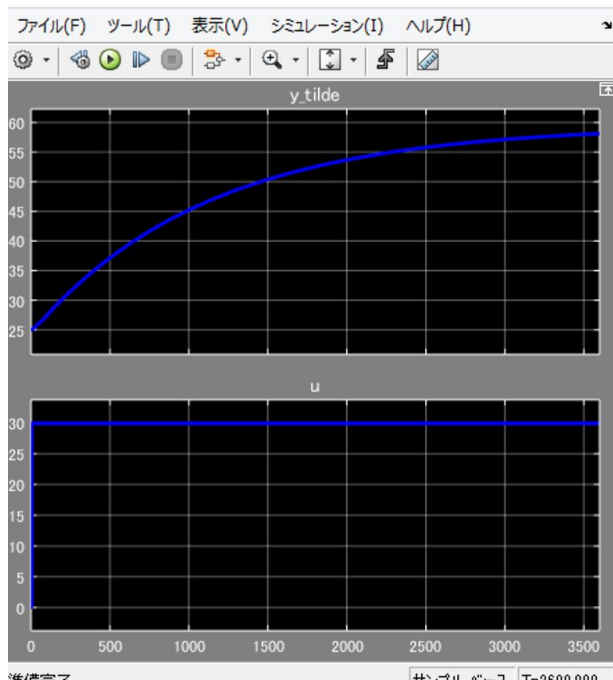


Example Output

Heater :30W
input

Start :25°C

Final :59°C



Caution! Do not use large input for step test

but why?

- Does large input damage the plant ?
-

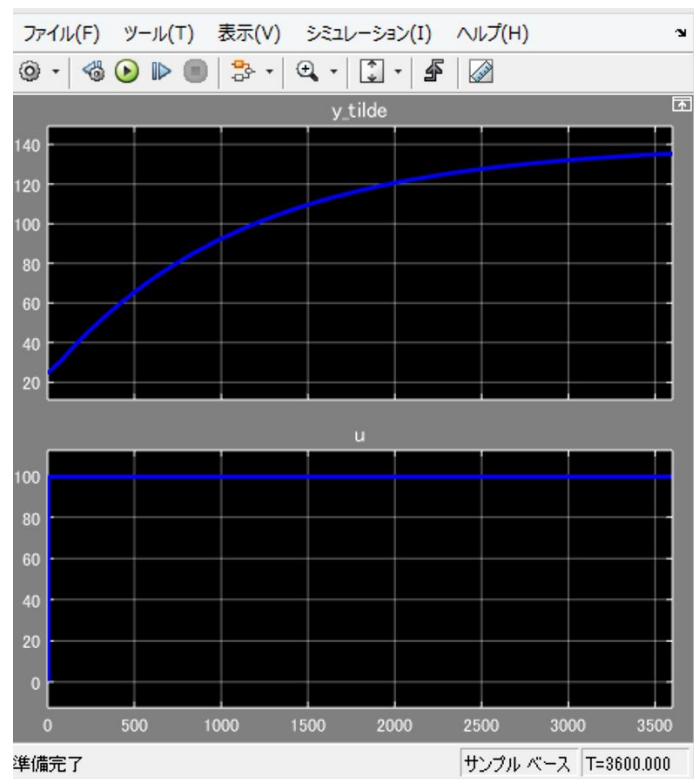
Caution! Do not use large input for step test

Heater Input: 100W

Start: 25°C

Final: 135°C

What's the problem?



Caution! Do not use large input for step test

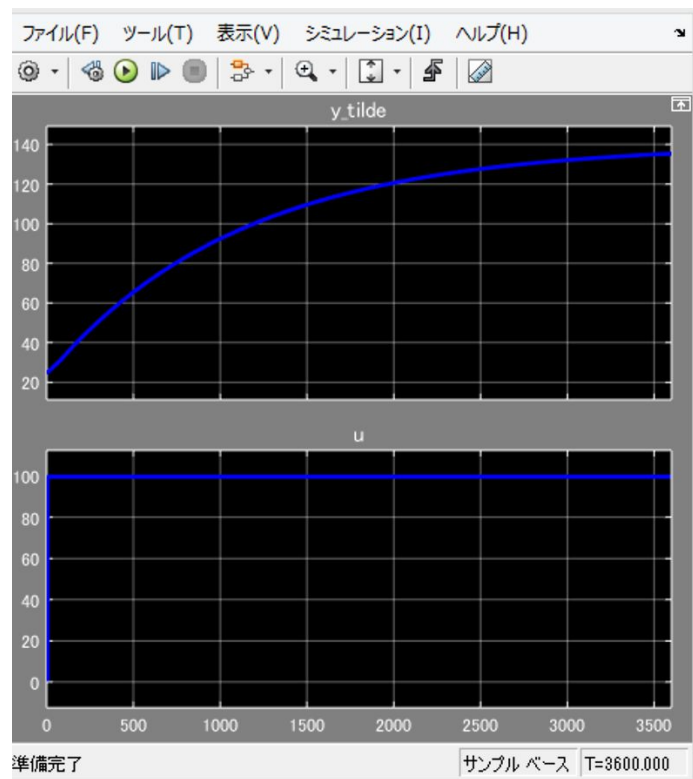
Heater Input: 100W

Start: 25°C

Final: 135°C

This model does not include **boiling phenomenon**.

We can not measure final temperature as a result.



For advanced study

<https://jp.mathworks.com/help/ident/examples/estimating-continuous-time-models-using-simulink-data.html>

ヘルプ センター

サポートを検索する

サポート

目次

ドキュメンテーションのホーム

System Identification Toolbox

線形モデルの同定

Simulink データを使用した連続時間モデルの推定

項目一覧

Simulink モデルからのシミュレーションデータの取得

シミュレーション データを使用した離散モデルの推定

推定の改良

離散時間モデルの連続時間モデルへの変換 (LTI)

連続時間モデルの直接推定

不確かさの解析

連続時間推定におけるサンプル間動作の考慮

その他

ビデオ

MATLAB Answers

評価版

製品の更新

このページは前リリースの情報です。該当の英語のページはこのリリースで削除されています。

Simulink データを使用した連続時間モデルの推定

R2020a

この例では、System Identification Toolbox™ を使用して、Simulink® でシミュレートされたモデルを同定する方法を説明します。この例では、連続時間システムおよび遅延を処理する方法と、入力サンプル間動作の重要性について説明します。

この例では次を使用します:

System Identification Toolbox

Simulink

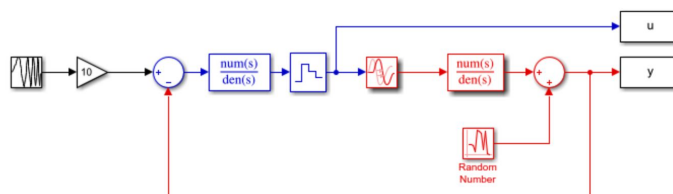
View MATLAB Command

```
if exist('start_simulink','file')~=2
    disp('This example requires Simulink.')
    return
end
```

Simulink モデルからのシミュレーション データの取得

以下の Simulink モデルで記述されるシステムを見てみましょう。

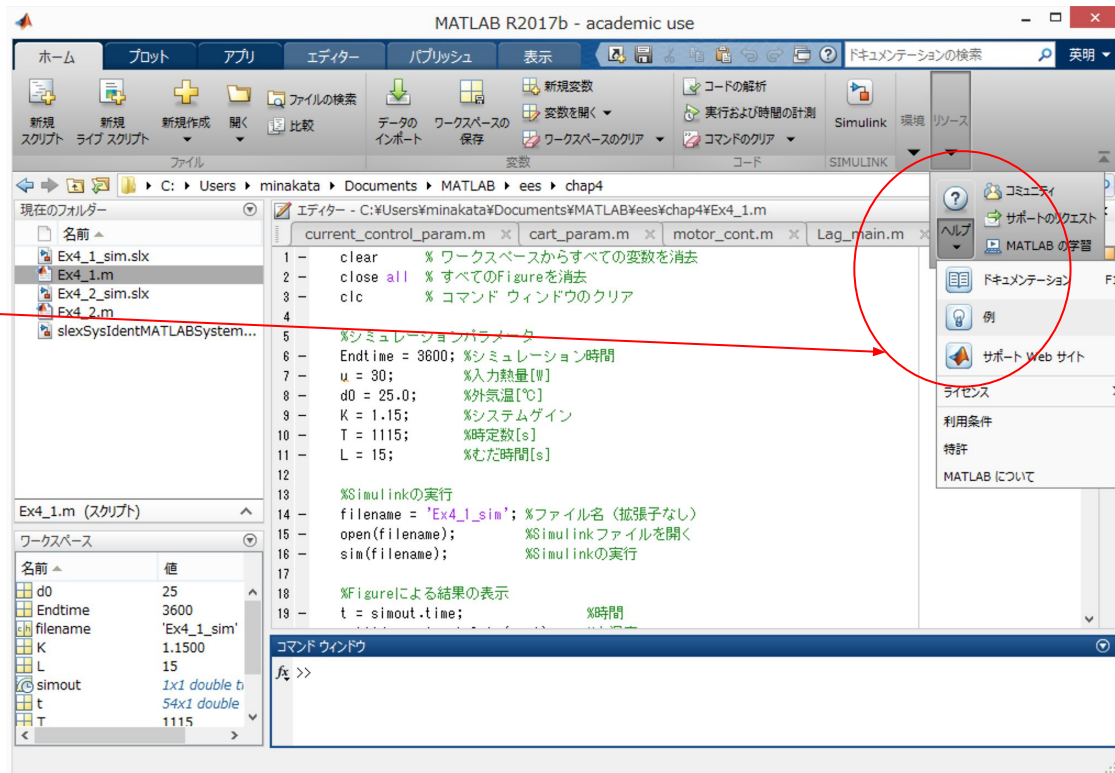
```
open_system('iddems1')
set_param('iddems1/Random Number','seed','0')
```



An Example of Digital Filter System Identification

Open from MATLAB
Window

help->demo
(ヘルプ->例)



An Example of Digital Filter System Identification

Search "Identification"
or "同定"

Select "demo" or "例"

Choose this sample.



An Example of Digital Filter System Identification

Open this model.

The screenshot displays the MATLAB System Block Example: System Identification for an FIR System using LMS Adaptive Filtering. The window is titled "ヘルプ" (Help) and shows the MATLAB System Block Example: System Identification for an FIR System using LMS Adaptive Filtering. The left sidebar contains a table of contents with the following items:

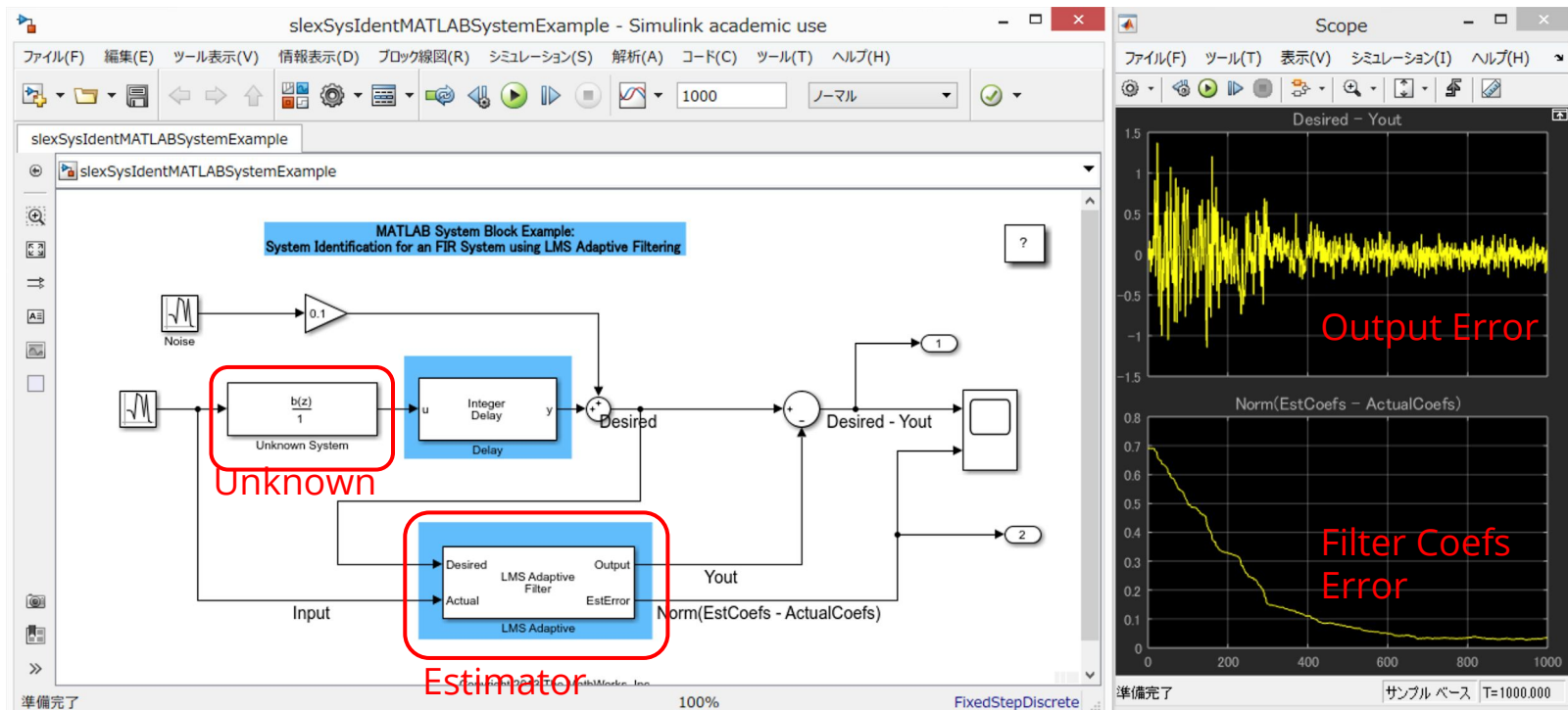
- 例のホーム
- Simulink
 - モデリング機能
 - S-Function、System object および ガシ コード ツールによるカスタム ブロック
- Simulink
 - ブロックの作成
 - System object の統合
- MATLAB System ブロックを使用した FIR システムのシステム同定
 - 項目一覧

The main content area shows the title "MATLAB System ブロックを使用した FIR システムのシステム同定" and a description of the example. It includes a "モデルを開く" (Open Model) button and a list of bullet points explaining the components and the LMS Adaptive Filter block.

MATLAB System Block Example: System Identification for an FIR System using LMS Adaptive Filtering

The diagram illustrates the system identification process. It starts with a "Random Number" block and a "Noise" block. The "Random Number" block is connected to an "Unknown System" block. The "Noise" block is connected to a gain block of 0.1, which is then connected to the "Unknown System" block. The output of the "Unknown System" block is labeled "u". This signal "u" is fed into an "Integer Delay" block, which outputs "y". The signal "y" is also fed into a "Desired" block. The output of the "Desired" block is labeled "Desired". The signal "Desired" is then fed into an "LMS Adaptive Filter" block, which outputs "Output". The signal "Output" is fed into a "Scope" block. The signal "Output" is also fed into a "Desired - Yout" block, which outputs "Out2". The signal "Out2" is fed into a "Scope" block. The signal "Out2" is also fed into a "Desired" block, which outputs "Out1".

Output of this sample



Filters Coefs

Unknown

Estimator

The top window shows the variable 'b' with the following values for indices 1 to 10:

Index	1	2	3	4	5	6	7	8	9	10
1	-0.0012	-0.0014	0.0020	0.0029	-0.0044	-0.0064	0.0090	0.0125	-0.0169	-0.0200

The bottom window shows the variable 'a' with the following values for indices 1 to 10:

Index	1	2	3	4	5	6	7	8	9	10
1	0	-0.0012	-0.0014	0.0020	0.0029	-0.0044	-0.0064	0.0090	0.0125	-0.0169